Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

AI and Application Development and Jira

M1:Practical – VII(D&E)

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Class: TYIT	Batch: 1
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4. Implement k-fold cross-validation on a classification model.

Code:

```
from sklearn.datasets import load_iris

from sklearn.model_selection import cross_val_score

from sklearn.neighbors import KNeighborsClassifier

#Load Dataset

iris=load_iris()

X=iris.data

y=iris.target

#Create classification model

model=KNeighborsClassifier()

#Apply 5-fold cross-validation

scores=cross_val_score(model,X,y,cv=5)

print("Accuracy scores for each fold:")

print(scores)

print("Average Accuracy:",
round(scores.mean()*100,2),"%")

print("Asmitha T002")
```

Output:

```
Accuracy scores for each fold:
[0.96666667 1.         0.93333333 0.96666667 1.        ]
Average Accuracy: 97.33 %
Asmitha T002
```

With Different Dataset:

Code:

```
from sklearn.datasets import load_breast_cancer

from sklearn.model_selection import cross_val_score

from sklearn.neighbors import KNeighborsClassifier

#Load Dataset

Breast_Cancer=load_breast_cancer()

X=Breast_Cancer.data

y=Breast_Cancer.target

#Create classification model

model=KNeighborsClassifier()

#Apply 5-fold cross-validation

scores=cross_val_score(model,X,y,cv=5)

print("Accuracy scores for each fold:")

print(scores)

print("Average Accuracy:",
round(scores.mean()*100,2),"%")

print("Asmitha T002")
```

Output:

```
Accuracy scores for each fold:
[0.88596491 0.93859649 0.93859649 0.94736842 0.92920354]
Average Accuracy: 92.79 %
Asmitha T002
```

Dataset Comparison :

Iris	Breast Cancer
150 samples	569 samples
4 features	30 features
3 classes	2 classes
Simple dataset	Complex dataset

Justification:

- **Iris is small and easy, suitable for learning.**
- **Breast Cancer is larger and more realistic, giving a more reliable evaluation.**
- **KNN performs well on both datasets.**

Conclusion: The Breast Cancer dataset is better for evaluating model performance because it is larger and more representative of real-world data.

5. Implement a Decision Tree classifier using the Iris or any relevant dataset.

Code:

```
from sklearn.datasets import load_iris

from sklearn.model_selection import
train_test_split

from sklearn.tree import DecisionTreeClassifier

#Load dataset

iris=load_iris()

X=iris.data

y=iris.target

#Split dataset

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.3,
    random_state=1
)

#Create model

model=DecisionTreeClassifier()

#Train model

model.fit(X_train, y_train)

# Display accuracy

print("Training Accuracy:", model.score(X_train,
y_train) * 100)

print("Testing Accuracy:", model.score(X_test,
y_test) * 100)

# Prediction

prediction = model.predict([X_test[0]])

print("Predicted Class:", prediction[0])

print("Asmitha T002")
```

Output:

```
Training Accuracy: 100.0
Testing Accuracy: 95.55555555555556
Predicted Class: 0
Asmitha T002
```

With Different Dataset:

Code:

```
from sklearn.datasets import load_wine

from sklearn.model_selection import
train_test_split

from sklearn.tree import DecisionTreeClassifier

#Load dataset

wine=load_wine()

X=wine.data

y=wine.target

#Split dataset

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.3,
    random_state=1
)

#Create model

model=DecisionTreeClassifier()

#Train model

model.fit(X_train, y_train)

# Display accuracy

print("Training Accuracy:", model.score(X_train,
y_train) * 100)

print("Testing Accuracy:", model.score(X_test,
y_test) * 100)

# Prediction

prediction = model.predict([X_test[0]])

print("Predicted Class:", prediction[0])

print("Asmitha T002")
```

Output:

```
Training Accuracy: 100.0
Testing Accuracy: 92.5925925925926
Predicted Class: 2
Asmitha T002
```

Dataset Comparison :

Iris	Wine
150 samples	**178 samples**
4 features	**13 features**
3 classes	**3 classes**
Simple dataset	**More detailed dataset**

Justification

- **Iris is simple and suitable for learning Decision Trees.**
- **Wine has more features, making classification more challenging and realistic.**
- **Decision Tree performs well on both datasets.**

Conclusion: The Wine dataset is better for evaluating the Decision Tree classifier because it contains more features and represents a more complex classification problem.